Customer Shopping Behavior Analysis

1. Project Overview

In this project, I analyzed customer shopping data to understand how customers behave while purchasing products.

The main goal of this project is to find useful insights like spending patterns, popular products, and customer types. This helps businesses make better decisions.

2. Dataset Summary

Total records: 3,900
Total columns: 18
The dataset includes:

- Customer details (age, gender, location, subscription)
- Purchase details (product, category, amount, season, size, color)
- Shopping behavior (discount, purchase frequency, ratings, shipping type)
- There were some missing values in the review rating column.

3. . Data Cleaning & Preprocessing

I used Python to clean and prepare the data:

- **Loaded data using pandas: -** Imported the dataset using pandas.

- **Checked data structure and summary: -** Used `df.info()` to check structure and `df.describe()` for summary statistics.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

- **Handled missing values: -**

- I checked for missing values In dataset.
- The **Review Rating** column had Some missing values.
- I filled these missing values using the **mean value** of each product category to maintain accuracy.

❖ **Column Renaming: -**

Renamed column names into **snake_case** format.

This improved readability and made it easier to use in coding.

❖ **Created new columns: -**

- Created **age_group** column by binning customer ages.
- Created **purchase_frequency_days** column from purchase data.

❖ **Data consistency Check: -**

Checked if some columns had duplicate of similar information.

Found that **promo_code_used** was similar **to discount_applied** .

Removed unnecessary column to simplify the dataset.

After cleaning, I stored the data in a MYSQL database for further analysis.

4. Data Analysis using SQL

I used SQL to answer important business questions:

1. **Revenue by Gender** – Compared total revenue generated by male vs. female customers.

	gender	SUM(purchase_amount)
▶	Male	157890
	Female	75191

2. **High-Spending Discount Users** – Identified customers who used discounts but still spent above the average purchase amount.

customer_id	purchase_amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79

customer 2

3. **Top 5 Products by Rating** – Found products with the highest average review ratings.

item_purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.78

4. **Discount-Dependent Products** – Identified 5 products with the highest percentage of discounted purchases.

	item_purchased	discount_rate
▶	Hat	50.00
	Sneakers	49.66
	Coat	49.07
	Sweater	48.17
	Pants	47.37

5. **Customer Segmentation** – Classified customers into New, Returning, and Loyal segments based on purchase history.

	customer_segment	Number Of Customer
▶	Loyal	3116
	Return	701
	New	83

6. **Top 3 Products per Category** – Listed the most purchased products within each category.

	item_rank	category	item_purchased	total_orders
▶	1	Accessories	Jewel Jewelry	171
	2	Accessories	Sunglasses	161
	3	Accessories	Belt	161
	1	Clothing	Blouse	171
	2	Clothing	Pants	171
	3	Clothing	Shirt	169
	1	Footwear	Sandals	160
	2	Footwear	Shoes	150
	3	Footwear	Sneakers	145
	1	Outerwear	Jacket	163
	2	Outerwear	Coat	161

7. **Revenue by Age Group** – Calculated total revenue contribution of each age group.

age_group	total_revenue
Young Adult	62143
Middle-Aged	59197
Adult	55978
Senior	55763

8. **Repeat Buyers & Subscriptions** – Checked whether customers with >5 purchases are more likely to subscribe.

subscription_status	COUNT(customer_id)
Yes	958
No	2518

5. Dashboard in Power BI

I created a dashboard in Power BI to visualize the data.

6. Business Recommendations

- Improve subscription benefits to attract more users
- Give rewards to loyal customers
- Use discounts carefully
- Promote top-rated products
- Target specific customer groups for marketing